

Inter-Process Communication

Inter-Process Communication (IPC) is a mechanism that allows two or more processes to communicate with each other and exchange data while executing in an operating system.

IPC techniques:

There are in total 8 IPC techniques in which are 2 as follows:

1.Semaphores:

A semaphore is a synchronization tool used by an operating system to manage access to shared resources. It uses wait and signal operations to coordinate processes and prevent race conditions.

Example: Think of a single key to a room. Only the person holding the key can enter. When that person leaves, they return the key so another person can enter.

The key represents the semaphore, and the room represents the shared resource.

In short: A semaphore prevents multiple processes from accessing a limited shared resource at the same time.

2. Signals:

A signal is a software notification sent to a process to inform it that a particular event has occurred. It is used by the operating system or another process to notify a process that it needs to respond to an event.

Example: Imagine students are attending a class when the fire alarm rings, The fire alarm rings to notify everyone that there is an emergency, Students receive the notification, They stop their current activity, They follow the instructions and leave the building safely.

This is similar to a signal in an operating system, where a process receives a notification about an event and respond to it.

